

A Theme Based Project Report

on

MIND MATE

Submitted in partial fulfilment of the requirements

of the degree of

Bachelor of Engineering

in

Computer Science and Engineering

(Artificial Intelligence and Machine Learning)

by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MATURI VENKATA SUBBA RAO ENGINEERING COLLEGE

(An Autonomous Institution, Sponsored by Matrusri Education Society - Estd. 1980)

Affiliated to Osmania University & Recognized by AICTE

Nadergul (PO), Balapur (M), Hyderabad, Telangana, India – 501510

Academic Year: 2023 – 2024



CERTIFICATE

*This is to certify that the Theme Based project work entitled “**Mind Mate**” is a bonafide work carried out by **K. Revanth Siddartha**(2451-22-748-017), **Mohammad Shoaib Alam**(2451-22-748-020), **Vuyyala Shivani**(2451-22-748-023) in partial fulfilment of the requirements for the award of degree of Bachelor of Engineering in Computer Science and Engineering (Artificial Intelligence and Machine Learning) from Maturi Venkata Subba Rao(MVSR) Engineering College, affiliated to OSMANIA UNIVERSITY, Hyderabad, during the Academic Year 2023-2024 under our guidance and supervision.*

The results embodied in this report have not been submitted to any other university or institute for the award of any degree or diploma.

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DECLARATION

This is to certify that the work reported in the present Theme Based project entitled “**Mind Mate**” is a record of bonafide work done by us in the Department of Computer Science and Engineering, MVSR Engineering College, Osmania University. The reports are based on the work done entirely by us and not copied from any other source. The results embodied in this report have not been submitted to any other University or Institute for the award of any degree or diploma to the best of our knowledge and belief.

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ABSTRACT

The shortage of mental health professionals, along with the high cost of therapy, has made it difficult for many people to access the care they need, Mind Mate is a mental health AI powered chat assistant which guides and helps you to improve your mental health. Mind Mate website provides a user-friendly UI to learn about mental health. This application provides a personalized mental wellness experience using AI. It's designed to offer emotional support and coping strategies through empathetic conversations, You can share your thoughts and feelings confidentially and without worry, anytime you want, and get feedback in a seemingly conscientious and thoughtful manner

CERTIFICATIONS



Statement of Achievement

kanaparthi Revanth Siddartha

has successfully achieved student level credential for completing the **JavaScript Essentials 1** course, provided by Cisco Networking Academy in collaboration with OpenEDG JS Institute.

The graduate is able to proficiently:

- Understand the syntax of the core JavaScript language that allows for working with variables, operators, flow control, and functions.
- Understand the basics of the JavaScript data types system, distinguishing between primitive and complex types.
- Think algorithmically and can analyze a problem using a programmatic conceptual apparatus.
- Choose a data type adequate to the problem being solved and use suitable flow control means.
- Design, develop, and improve very simple JavaScript programs.
- Interpret and handle basic exceptions related to errors in program execution.
- Understand a programmer's work in the software development process and the role of fundamental development tools.
- Understand how a program is interpreted and executed in an actual computer environment, local or remote.



June 07, 2024

Laura Quintana
Laura Quintana
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Cisco Networking Academy



Statement of Achievement

Mohammad Shoaib Alam

has successfully achieved student level credential for completing the **JavaScript Essentials 1** course, provided by Cisco Networking Academy in collaboration with OpenEDG JS Institute.

The graduate is able to proficiently:

- Understand the syntax of the core JavaScript language that allows for working with variables, operators, flow control, and functions.
- Understand the basics of the JavaScript data types system, distinguishing between primitive and complex types.
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The graduate is able to proficiently:

- Understand the syntax of the core JavaScript language that allows for working with variables, operators, flow control, and functions.
- Understand the basics of the JavaScript data types system, distinguishing between primitive and complex types.
- Think algorithmically and can analyze a problem using a programmatic conceptual apparatus.
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June 07, 2024

Laura Quintana
Laura Quintana
Vice President and General Manager
Cisco Networking Academy

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Vision

To impart technical education of the highest standards, producing competent and confident engineers with an ability to use computer science knowledge to solve societal problems.

Mission

- To make the learning process exciting, stimulating, and interesting.
- To impart adequate fundamental knowledge and soft skills to students.
- To expose students to advanced computer technologies to excel in engineering practices by bringing out the creativity in students.
- To develop economically feasible and socially acceptable software.

Program Educational Objectives (PEOs)

The Bachelor's program in Computer Science and Engineering is aimed at preparing graduates who will: -

PEO-1: Achieve recognition through demonstration of technical competence for successful execution of software projects to meet customer business objectives.

PEO-2: Practice life-long learning by pursuing professional certifications, higher education, or research in the emerging areas of information processing and intelligent systems at a global level.

PEO-3: Contribute to society by understanding the impact of computing using a multidisciplinary and ethical approach.

Program Outcomes (POs)

PO 1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization for the solution of complex engineering problems.

PO 2: Problem analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with

appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and the need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions

PO 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOS)

PSO-1: Demonstrate competence to build effective solutions for computational real-world problems using software and hardware across multi-disciplinary domains.

PSO-2: Adapt to current computing trends for meeting the industrial and societal needs through a holistic professional development leading to pioneering careers or entrepreneurship.

COURSE OBJECTIVES AND OUTCOMES

Course Title: Theme Based Project

Course Code: U22PW481AL

Course Objectives

- To enhance practical and professional skills.
- To familiarize tools and techniques of systematic literature survey and documentation.
- To expose the students to industry practices and teamwork.
- To encourage students to work with innovative and entrepreneurial ideas.

Course Outcomes

- Demonstrate the ability to synthesize and apply the knowledge and skills acquired in the academic program to the real world problems.
- Evaluate different solutions based on economic and technical feasibility
- Effectively plan a project and confidently perform all aspects of project management.

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CHAPTER - 1

INTRODUCTION

Mind Mate is an innovative AI-powered chatbot designed to tackle the challenges of accessing mental health care due to the shortage of professionals and high therapy costs. This chatbot provides personalized mental wellness support, offering empathetic and confidential conversations to help users improve their mental health. The Mind Mate Project is a web application designed to improve mental health care by using modern technology and proven therapy methods. It aims to provide easy and personalized mental health support through an online platform. By using artificial intelligence, Mind Mate analyses user data to offer tailored help, real-time feedback, coping strategies, and access to professional resources. The project focuses on being private, easy to use, and effective, aiming to reduce the stigma around mental health treatment and make support more accessible. The ultimate goal of Mind Mate is to improve overall well-being, build resilience, and encourage proactive mental health care.

1.1 Problem statement

Design an conversational AI capable of providing empathetic, personalized, and effective mental health support to users in need.

1.2 Existing systems

Individuals primarily relied on traditional face-to-face therapy sessions with licensed mental health professionals, such as psychologists, psychiatrists, counselors, or therapists. These sessions typically involved scheduling appointments and attending in-person meetings, which present several challenges. Traditional therapy can be hard to access because it is often expensive, requiring significant out-of-pocket expenses or comprehensive insurance coverage. Additionally, there can be long wait times to get an appointment, sometimes taking weeks or even months. For those living in remote or underserved areas, accessing in-person therapy can be difficult due to geographical barriers.

1.3 Proposed System

The proposed Mind Mate web application aims to revolutionize mental health care by addressing the limitations of traditional therapy. This innovative system leverages modern technology to provide accessible and personalized mental health support. Users can access the platform from anywhere with an internet connection, eliminating geographical barriers. Through artificial intelligence and machine learning, Mind Mate analyses user data to offer tailored interventions, real-time

feedback, coping strategies, and access to professional resources. They offer convenience, privacy, and accessibility, making it easier for individuals to seek help and support for their mental health concerns without the barriers of traditional therapy.

1.4 Scope

The Mind Mate project leverages advanced web technology to revolutionize mental health care. By integrating artificial intelligence it offers personalized interventions, real-time feedback, and access to resources, aiming to enhance accessibility and reduce stigma associated with seeking help. This initiative empowers individuals to proactively manage their mental well-being through tailored support, marking a significant step forward in modern mental health care delivery.

CHAPTER - 2

2. SYSTEM ARCHITECTURE SPECIFICATION

2.1 Software Requirements

- IDE Used : Visual Studio
- Frontend : HTML, CSS, JavaScript
- Backend : Node js
- Database: MongoDB , Firebase

2.1.1 Functional Requirements

1. User Authentication

- Users can create an account and log in.

2. Chatbot Interaction

- Users can interact with the AI chatbot via text-based conversations.
- The chatbot will provide real-time responses using NLP algorithms.
- The chatbot will store and analyze chat logs to assess mood and sentiment.

3. Analysis

- The application will analyze user inputs to determine Mental Health using NLP.
- The application will store data in the Database.
- The application will perform time series analysis for the input data.

4. Weekly Report Generation

- The application will generate a weekly report summarizing the chat.
- The report will include observations, concerns recommendations for improvement.

- The report will be given based on the analysis when it is required for the user.

2.1.2 Non Functional requirements

Performance

- The application will provide real-time responses to user inputs.
- The application will generate weekly reports within a reasonable time frame.

Security

- The application will ensure user data privacy and security, employing encryption for data storage and transmission.

Usability

- The application will have a user-friendly interface that is easy to navigate.
- The application will provide clear instructions and feedback to users.

2.2 Hardware Requirements

- RAM: 8.00GB
- Processor: 11th Gen Intel ® Core™ i5-11320H @ 3.20GHz 2.50GHz
- Storage: 458GB

2.3 SYSTEM ARCHITECTURE

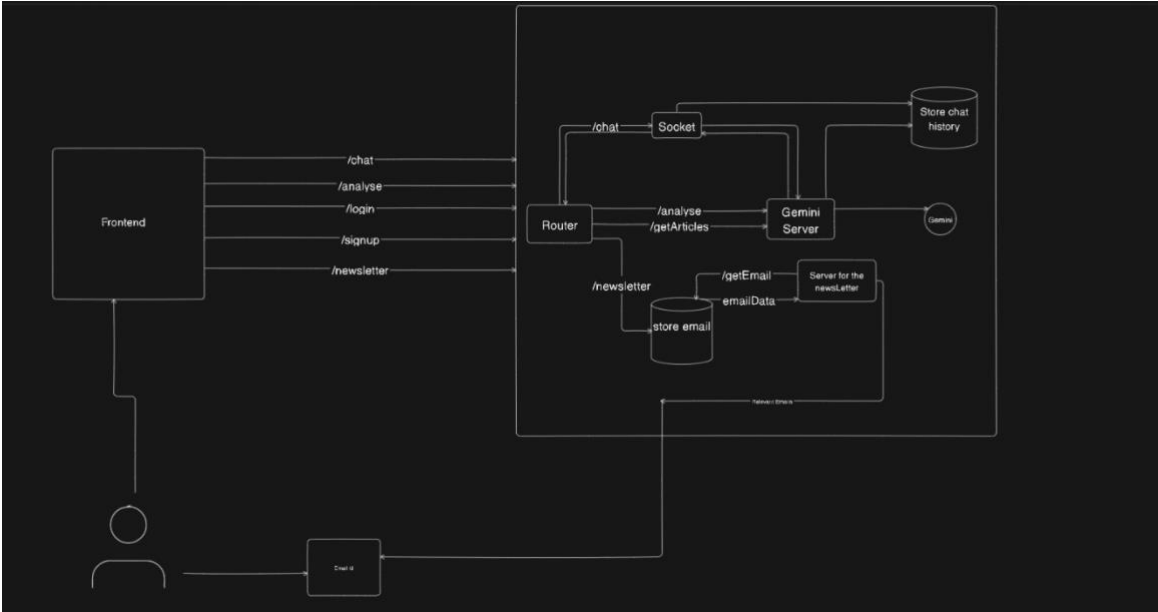


Fig 1.1 System Architecture

CHAPTER - 3

3. System Design

The objective of the system design phase is to create a comprehensive architecture and detailed components for the Mind Mate system. This phase involves defining the overall system architecture, which includes the client-server model and the database design. The design process also encompasses creating UI/UX wireframes for the frontend to ensure a user-friendly interface. Additionally, API specifications for backend services are outlined to establish clear communication protocols between the frontend and backend. All design elements are compiled into a detailed design document, providing a blueprint for the subsequent implementation phase.

3.1 Methodology

The methodology for the Mind Mate project involves a series of steps to ensure the development and deployment of a robust, efficient, and user-friendly mental health assistance platform. The process can be broken down into the following stages:

Implementation

The implementation phase focuses on developing the system components according to the design specifications. The frontend development begins with setting up the React environment, followed by the implementation of UI components based on the design wireframes. Firebase is integrated to handle authentication and provide real-time data updates, ensuring a seamless user experience. For the backend development, the Node.js environment is set up to build the server-side functionalities. RESTful APIs are developed to manage user requests efficiently. The email server is integrated to facilitate communication features, and database schemas are implemented in conjunction with Firebase to manage data storage and retrieval.

3.1.1 Algorithms Description

Natural Language Processing (NLP)

Mind Mate utilizes Natural Language Processing (NLP) to analyze and understand user input, facilitating personalized mental health support through empathetic and confidential conversations.

- **Text Classification:** Mind Mate employs text classification algorithms to categorize user messages into relevant mental health categories such as anxiety, depression, stress levels, and more. This helps in prioritizing and addressing user needs effectively.
- **Sentiment Analysis:** By leveraging sentiment analysis techniques, Mind Mate assesses the emotional tone of user messages to gauge their mental state and provide appropriate responses and interventions.
- **Named Entity Recognition (NER):** NER identifies entities such as mental health conditions, coping strategies, and emotional triggers mentioned in user texts. This enables Mind Mate to offer tailored resources and guidance based on identified needs.

Time-Series Analysis: Analyzes mental health data collected over time to identify trends, fluctuations, or patterns indicative of changes in mental health status. Visualizes this data using graphical representations such as line charts or histograms to provide users with insights into their mental well-being. Identify trends, periodic fluctuations, and anomalies in mental health metrics. Detect seasonal patterns (e.g., daily, weekly, seasonal variations) using statistical methods. Present insights through graphical representations such as line charts. Provide interactive features for users to explore and understand their mental health trends visually.

3.2 Structural Diagrams

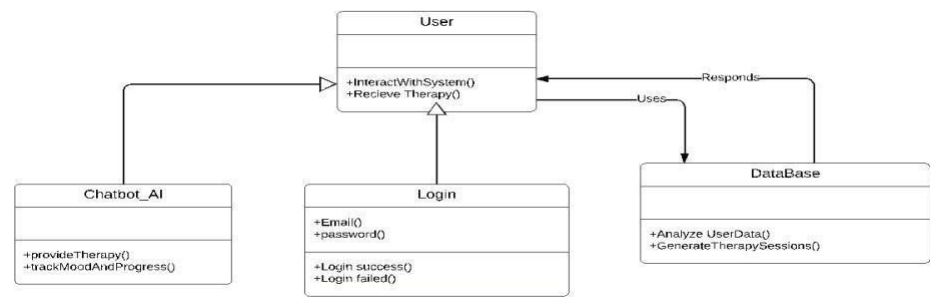


Fig 1.2 Class Diagram

3.3 Behavioral Diagrams

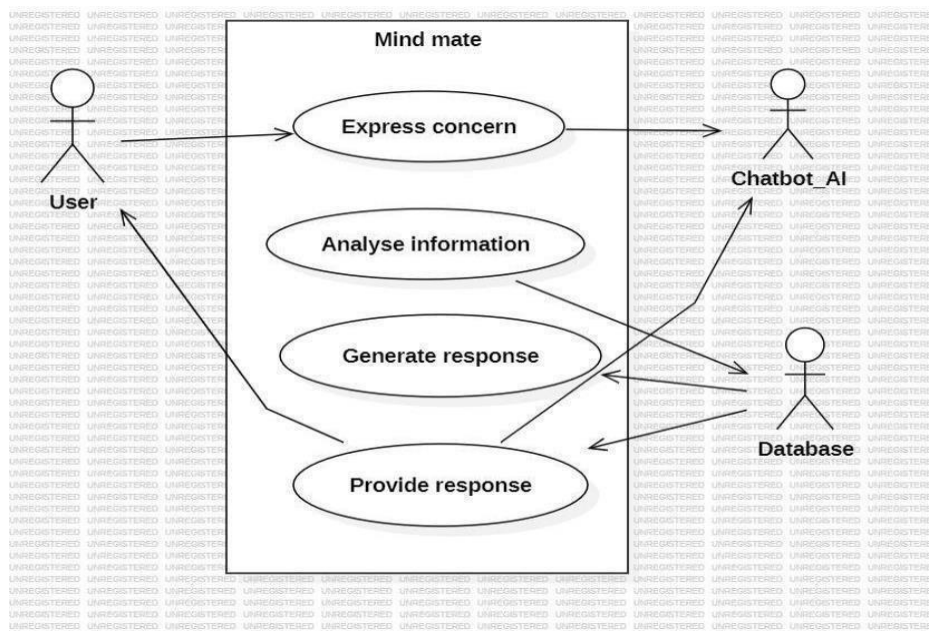


Fig 1.3 Use case Diagram

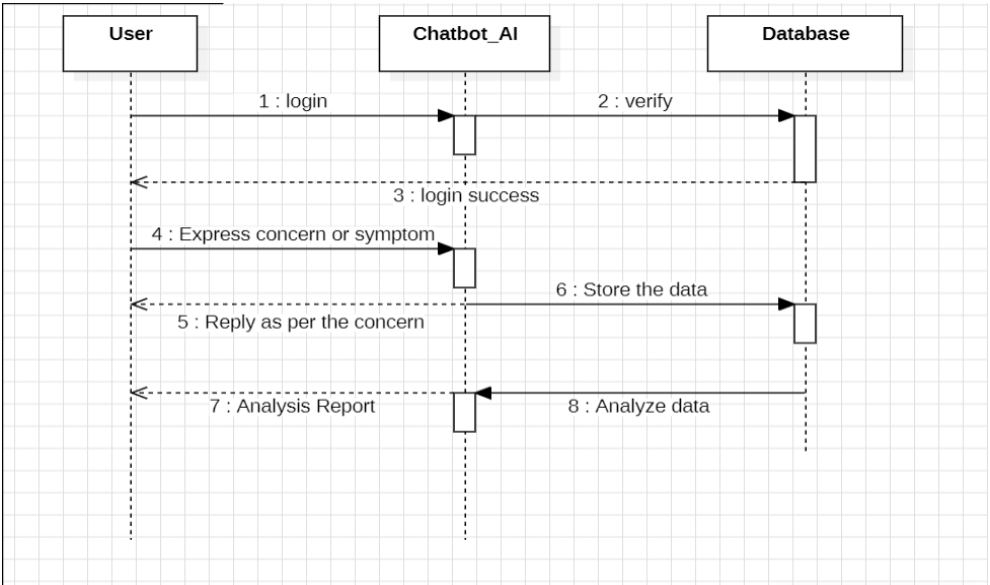


Fig 1.4 Sequence Diagram

CHAPTER - 4

4. Implementation

4.1 Environmental Setup

Installing an editor: Visual Studio

1. Download the Visual Studio by below mentioned link based on the operating system.

<https://code.visualstudio.com/download>

Download and Install MongoDB:

2. Visit the MongoDB Community Server download page

<https://www.mongodb.com/try/download/community>

Setting Up Node.js and npm

3. Download and Install Node.

<https://nodejs.org/en/>

Verify the installation by running 'node -v' and 'npm -v' in a terminal or command prompt.

Create a Firebase Project:

4. Go to the Firebase Console

<https://firebase.google.com/>

Add Firebase to your web app by following the setup instructions and copying the Firebase configuration code.

Mind Mate

Setting Up Gemini (Gen AI)

5.Integrate Gemini (Gen AI)

Follow the documentation provided by Gemini (Gen AI) to integrate their services into your project.

<https://ai.google.dev/gemini-api>

Ensure you have the necessary API keys and access credentials

4.2 Implementation Details

The Mind Mate project involves building an AI-powered mental health chatbot using modern web technologies, AI algorithms, and integrating it into a complete web application. The steps include frontend and backend development, database management and AI integration.

Frontend Development (React JS) Initialize

Project:

- Set up the React project using chat_react_app.

Create Components:

- Develop UI components such as chatBox, Dashboard and Profile.

Set Up Routing:

- Configure React Router for navigation between pages.

Integrate APIs:

- Use Axios or Fetch API to communicate with the backend.

Mind Mate

Styling:

- Style components using HTML and CSS-in-JS libraries.

Backend Development (Node.js)

Initialize Project:

- Set up a new Node.js project and install necessary dependencies.

Define Routes:

- Create route files for different functionalities like chatBox, Dashboard and Profile.

Create Services:

- Implement service functions for external API interactions using Gemini (Gen AI).

Define Database Models:

- Use Mongoose to define schemas and models for users, chats, and articles.

Set Up Middleware:

- Implement middleware for error handling, authentication, and logging.

Integrate WebSockets:

- Use the WS library to set up WebSocket communication for real-time chat.

Integrate Firebase Authentication:

- Use Firebase Authentication SDK to implement user sign-up and login.

Integrate API:

- Write service functions in the backend to interact with Gemini's API.

4.3 Results

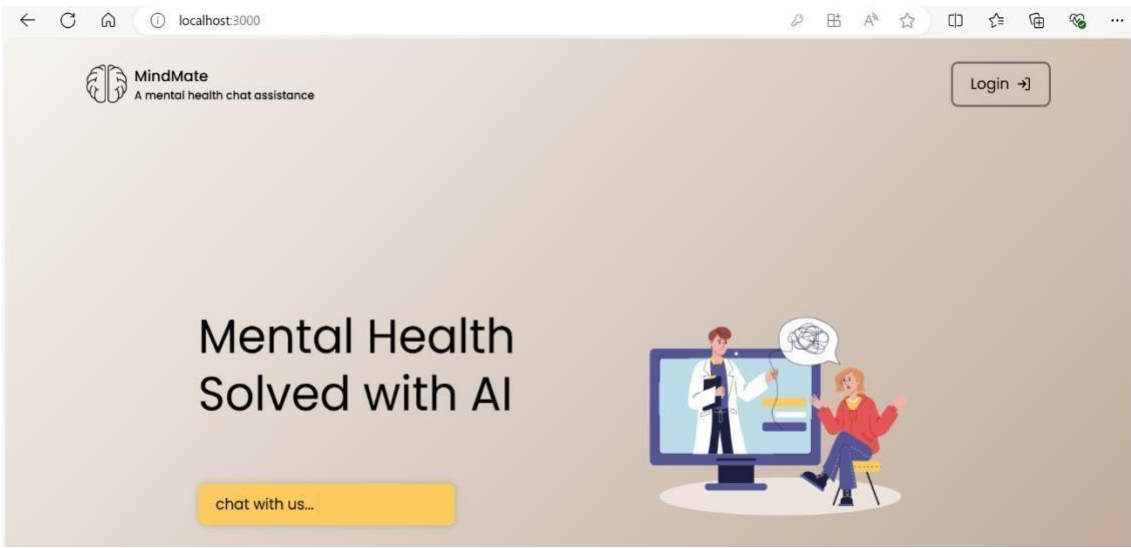


Fig 1.5 Home Page

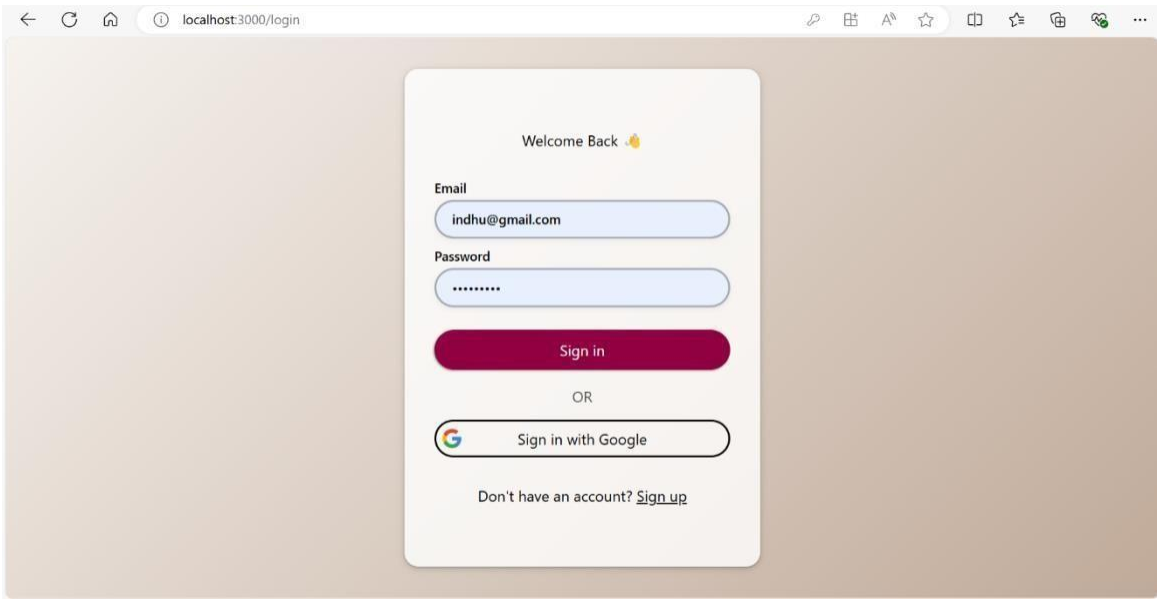


Fig 1.6 Login Page

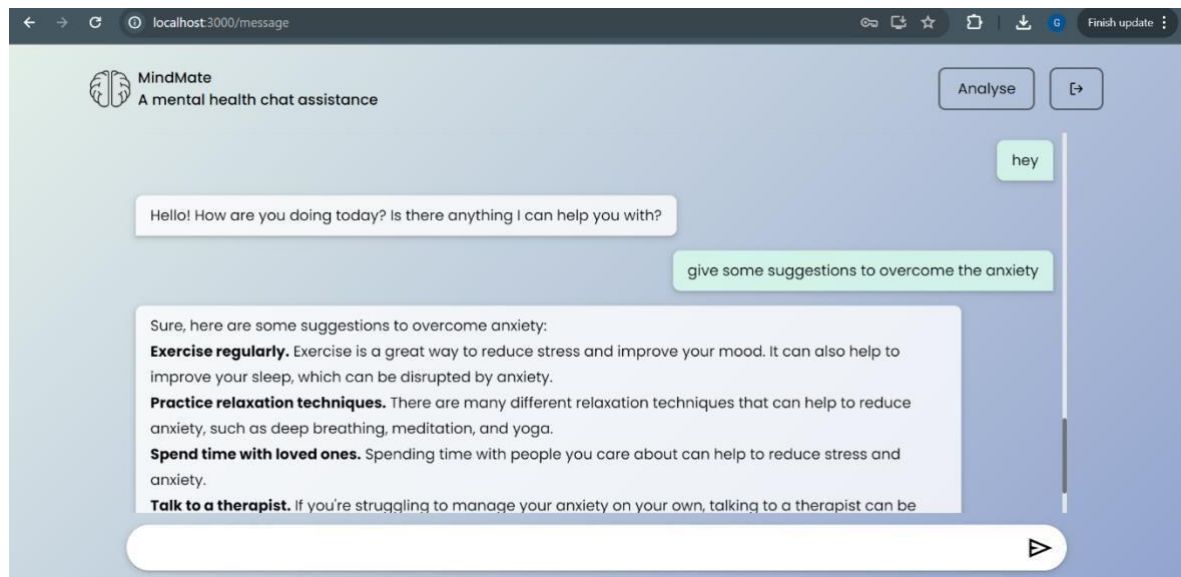


Fig 1.7 AI Chat BotTest

Cases

Test case	Description	Input	Output	Status
Login	User Login for Chat Assistant	Enter valid username and password	User is logged in and redirected to the dashboard	Pass
User Interaction	Initiate Mental Health Chat Session	Start a new chat session and interact with the AI	Chat session is initiated and AI provides guidance based on user inputs	Pass
	Initiate Mental Health Chat Session without Valid Input	Start a new chat session, provide nonsensical or invalid inputs (e.g., random characters or irrelevant text)	Error message indicating the input is not understood, or guidance to provide more relevant information	Pass
Analysis of Chat	Analyze Mental Health Over Time	Users previous chat sessions, navigate to the analysis page	Graphical representation of mental health trends over the period is displayed	Pass

CHAPTER-5

CONCLUSION & FUTURE ENHANCEMENTS

Conclusion

In a world where mental health resources are scarce and therapy is often financially out of reach, Mind mate emerges as a beacon of hope. Utilizing cutting-edge AI technology, Mind mate offers a lifeline to those in need of emotional support and guidance. With its intuitive user interface and empathetic conversations, Mind mate provides a safe space for individuals to explore their mental health journey confidentially.

Future Enhancements

In Future the Mind Mate project could include integrating YouTube suggestions and related articles to provide users with valuable resources and insights based on their interests. Enhancing email functionality with features like scheduling, templates, and signatures would improve user experience. Additionally, implementing real-time email tracking notifications and optimizing email management with advanced search and categorization options would further enhance the platform's capabilities.

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Appendix

Pseudo code

```
// Load necessary libraries and initialize configurations

loadLibraries()

initializeConfig()

// Define AI Model

model = loadGeminiModel(api_key)

// Function to handle user input and generate AI response

function handleUserInput(userInput):

    // Preprocess user input

    processedInput = preprocessInput(userInput)

    // Generate response using AI model

    response = model.generateResponse(processedInput)

    // Postprocess response if needed

    finalResponse =

    postprocessResponse(response) return

    finalResponse

// Function to preprocess user input

function preprocessInput(input):

    // Implement preprocessing steps like tokenization, normalization

    preprocessedInput = tokenize(input)
```

Mind Mate

```
    preprocessedInput = normalize(preprocessedInput) return  
  
    preprocessedInput  
  
// Function to postprocess AI response  
  
function postprocessResponse(response):  
  
    // Implement postprocessing steps if needed  
  
    postprocessedResponse = addFormatting(response) return  
  
    postprocessedResponse  
  
// Main loop to interact with the user while  
  
(userIsActive):  
  
    userInput = getUserInput()  
  
    aiResponse = handleUserInput(userInput)  
  
    displayResponse(aiResponse)
```